

PORTFOLIO

MUHAMMAD HUSEIN

Convosignum : Indonesian Sign Language App

Year Accomplished

Ongoing

Role/Position

UI/UX Designer

Project Type

Self-Initiated

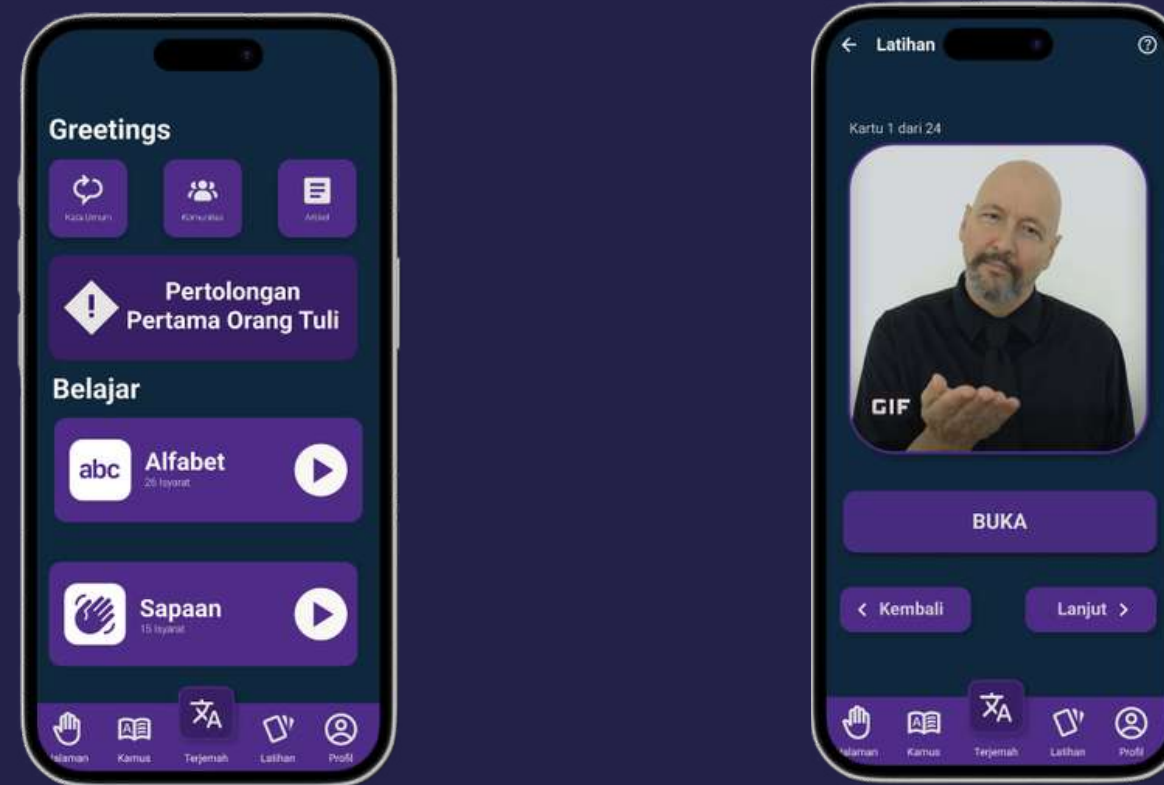
Publication Link

<https://www.figma.com/design/fMncnuHtR9wAM5MIArIKNG/Convosignum?node-id=0-1&t=1jZcAtgXMft2SuSk-1>

What I Learned

- Empathy helps design decision
- Create a design based on users pain point
- Learn how to apply design thinking to come up with a solution for users problem
- Explored innovative UI solutions to design a more intuitive and user-friendly experience.
- Learn to empathize with people whose daily life are very different to you

Convosignum, stands for *conversation* & *signum*, is an ongoing mobile app that bridge the gap between non-deaf and deaf people. This apps offer features like practice, image recognition for translation, and text-to-sign translation.



Why?

This project was motivated by my interest and fascination with the sign language and my desire to helps the struggle of communication faced by the deaf community. I notice there's not a lot of exposure for Indonesian deaf community on social media. I realized how isolating it must feel to live where people not understanding the struggle they have daily. And I feel like apart from being empathize towards them, I have to change and do something about it.

How?

I started with Design Thinking. Doing a lot of research about deaf community in Indonesia and BISINDO (Indonesian Sign Language), mapping pain points, and empathize before building the prototype.

The hardest challenge was designing for two very different users at once. A deaf person who needs transparency and understanding their struggles (pain point). And a hearing person who's a beginner and doesn't know where to start. I went through multiple iterations in Figma, scrapping early versions that felt too complex.

I'm optimistic that this concept can turn into an actual product with the with the existence of a sign language dataset and AI optimization.

Sentiment Analysis X Users Towards Apple Vision Pro using Naive Bayes Classifier

Year Accomplished
2024

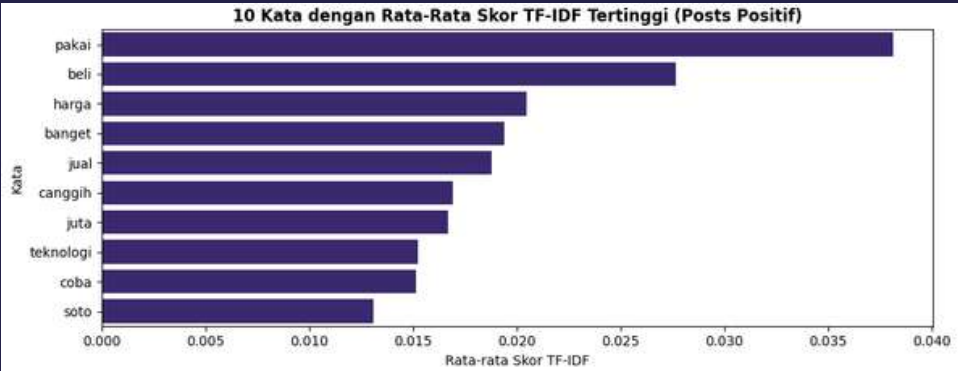
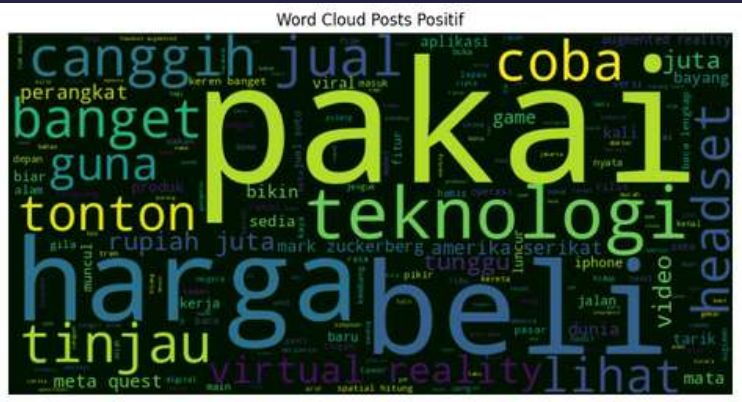
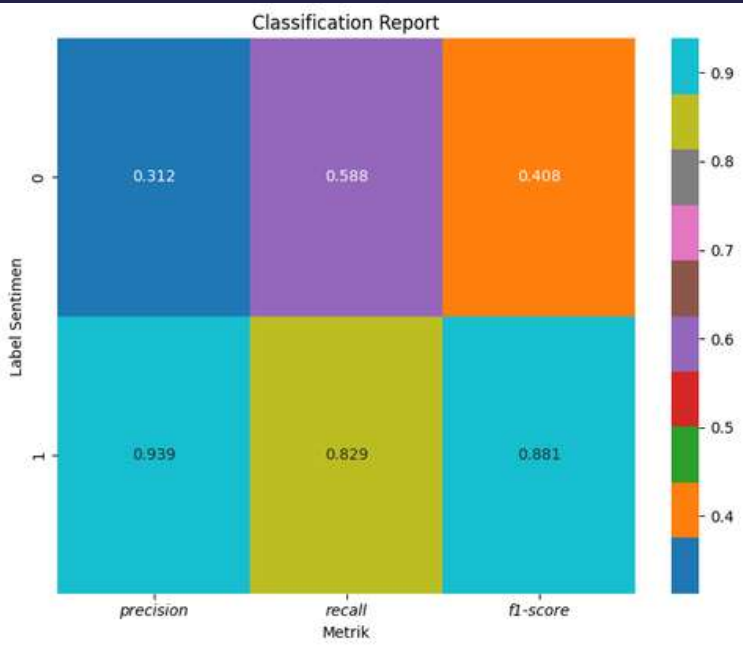
Role/Position
Developer and Researcher

Project Type
Self-Initiated (Undergraduate Thesis Project)

What I Learned

- Having an honest data is a key for reliable model and analysis.
- Understand the importance of preprocessing
- Handling an imbalanced dataset with SMOTE and saw the effect on model performance
- The importance of analysis visualization for storytelling and gaining better insights.
- The process of questioning my own results taught me more than just chasing for the high accuracy

Process of develop a model by analyzing the sentiment of social media users towards the product of Apple Vision Pro with Naive Bayes Classifier to understand how machine interprets Indonesian social media users perspective and feeling towards a trending up and coming product by determining if the posts are positive or negative.



How?

I cleaned 798 social media posts removing noise, symbols, inconsistent formatting and transformed text into numerical features using TF-IDF. **The challenges** were that there was a problem with the dataset being imbalanced. There are way too many Positive posts compared to negatives, which means my model tend to ignore negative posts and still getting decent scores. I don't like that because it's misleading. So I researched solutions and found SMOTE, an algorithm to handle imbalance dataset, so the model can learn both sides.

I got a high accuracy scores at first, but it was biased towards the positives one because of the imbalanced datasets, which got me thinking to explore it further. That quick thinking of not just accepting how it is, led to a more reliable model and a more honest analysis and ended up not losing too much accuracy score.

Why?

I wanted to understand what Indonesian users views on a new product Apple Vision Pro in which I thought was quite revolutionary. And I wanted to understand how machines interpret human emotion especially in Indonesian, which is full of slang, sarcasm, and mixed-language posts that even humans sometimes misread.

Coffee Shop Revenue Analysis

Linear Regression

Year Accomplished

2026

Role/Position

Data Analyst

Project Type

Self-Initiated

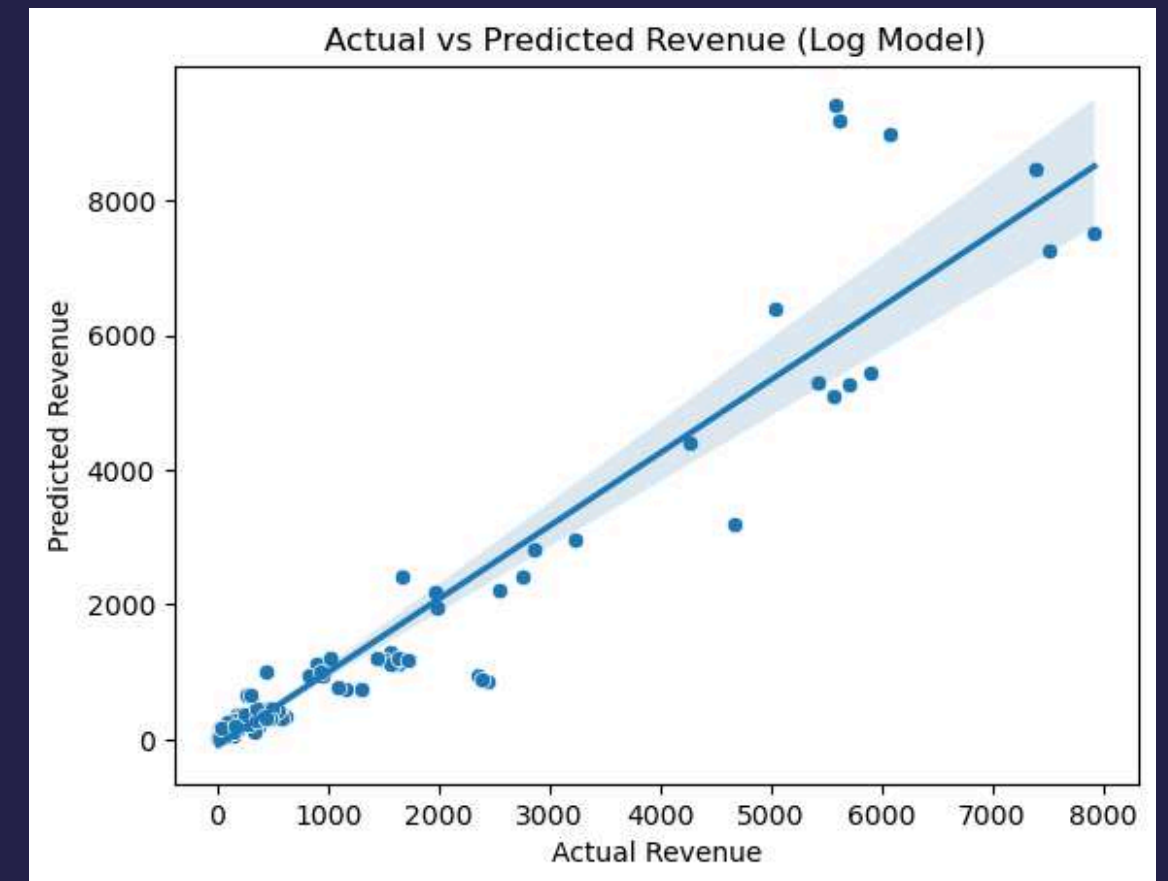
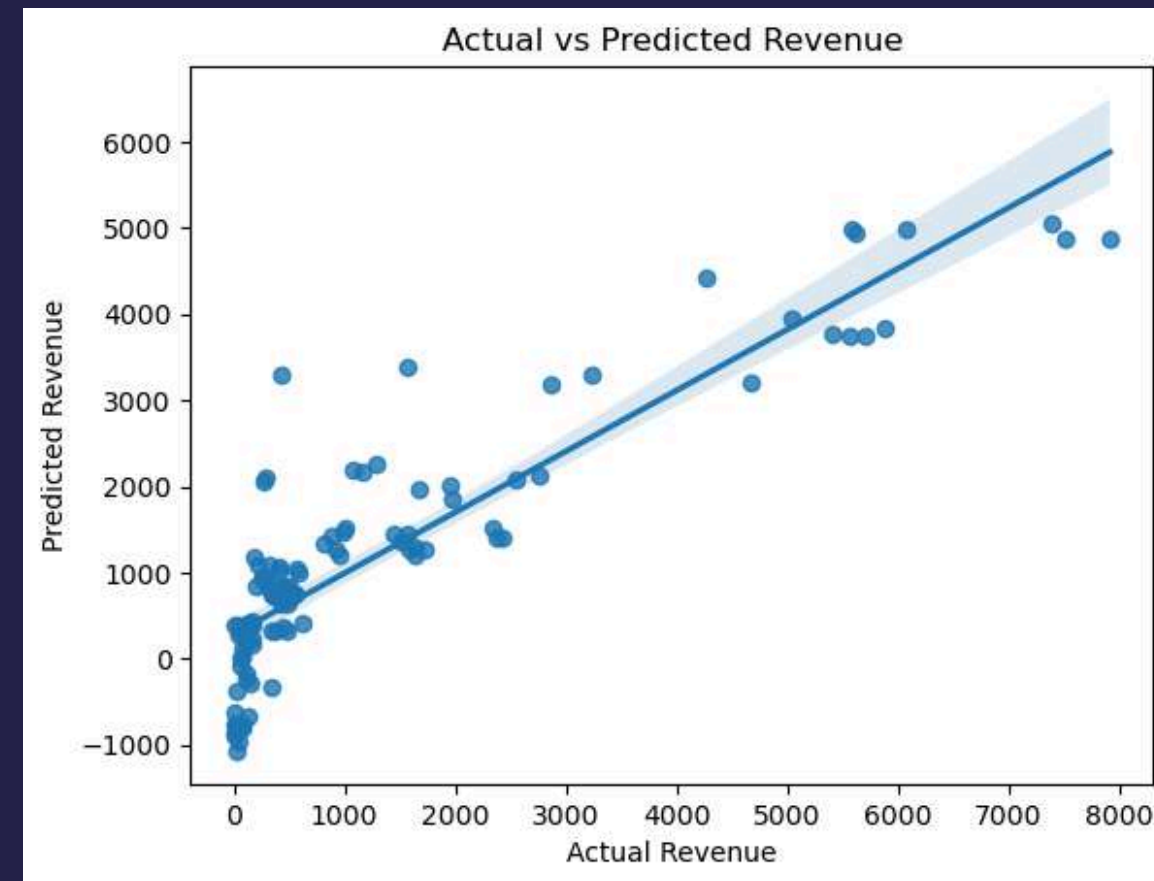
Publication Link

<https://github.com/fedexid/coffee-shop-mlr>

What I Learn?

- Learned the importance of feature engineering
- Testing OLS assumptions had a significant impact on the reliability of the results
- Log transformation can improve model stability leading to better regression performance
- Learn to approach problems analytically and iterate on models rather than relying on initial outcomes.

This is a Multiple Linear Regression Project as part of my progress of learning Data Science, with the goal to make model predictions that accurately will predict revenue based on factors like time period, product category, location of the store, and the day of the week. With also try to identify which factors that affects the model the most.



The process starts from exploratory data analysis and feature engineering on transactional data, and aggregating the revenue by summing it, I apply a multiple linear regression model and evaluated its performance. One of the main issue with base was heteroscedasticity which later I apply log transformation to better meet model assumptions and improve stability.

While the model achieved limited predictive performance, I learned that the analysis revealed patterns in how different factors influence revenue and highlighted the importance of OLS Assumptions in regression models.

Home Advantages in Football with Logistic Regression

Year Accomplished

2026

Role/Position

Data Science Developer

Project Type

Self-Initiated

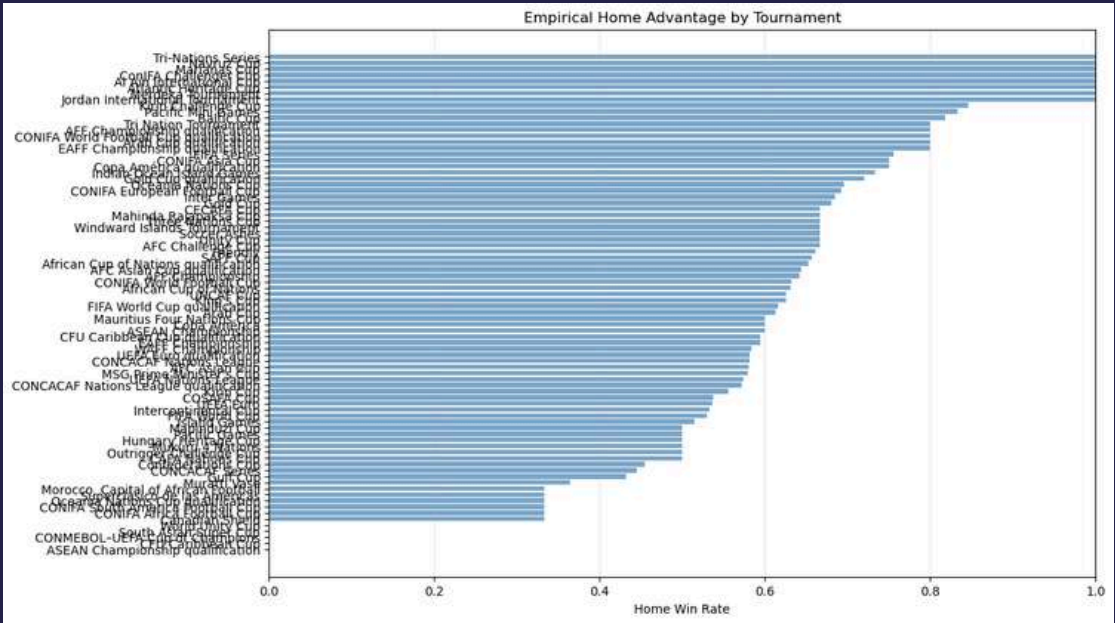
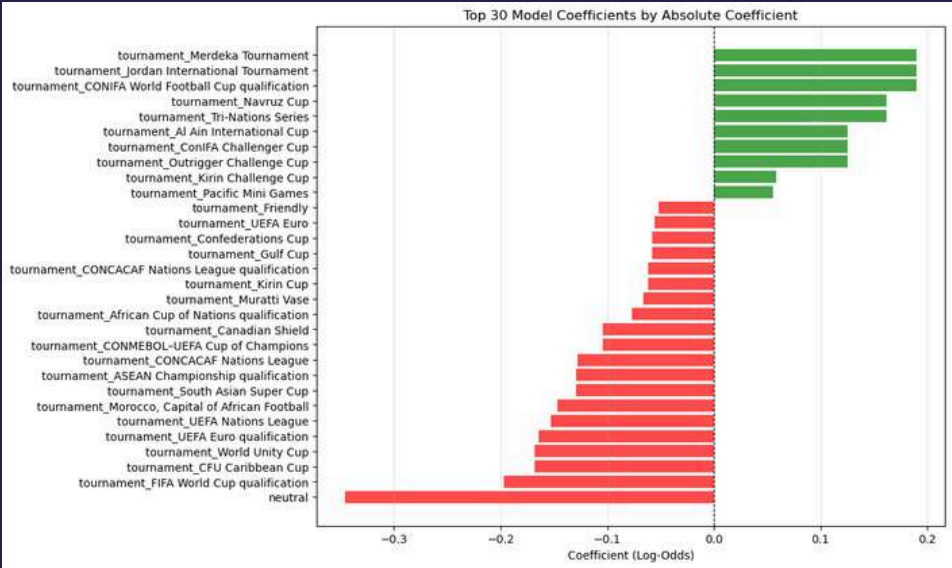
Publication Link

<https://github.com/fedexid/football-logistic>

What I Learned

- Building model is about decision not writing the best code
- Storytelling is half the work and the most important part, not the coding.
- Understanding football analytically with data helps me gain more knowledge to the game.
- Logistic regression is not just about yes or no answers, but the process showed me that there are underlying contexts and variables that explains the problems and outcomes.

This is a Multiple Linear Regression Project as part of my progress of learning Data Science, with the goal to make model predictions that accurately will predict the probability of home advantages in Football Matches for Football National Teams based on the 10 last years of international football match data (2016-2026)



How?

I use dataset of International Football Results from 1872 to 2026, but I only use the last 10 years. I built a Logistic Regression model to predict match outcomes based on whether a team is playing at home or neutral ground because I assume home advantage is not really a thing for most teams. The data cleaning process wasn't a challenge, The missing values are less than 1% so I removed them. I also remove results that ended in draw so I only focus on Win and Losses.

The real challenge was the decision making, every preprocessing choice affect the final model. so it's important not to overestimate it, because football results is very unpredictable due to factors like red cards, individual performances, etc.

Data showed that when home team playing at their own ground is an advantage in international matches. When the home team playing at neutral ground, it can reduce the home teams odds to win. Tournament type is also a factor, different tournaments show different home advantage rates.

Why?

As someone who loves to watch football analytically and currently learning data sciences on the side, I try to see the connection between these two field. A common saying in football is the “home advantages” gets better results. But I keep questioning it whether is real or a narrative that comes from coincidence.

World Cup is happening soon, and USA (who famously not one of the top teams in international world) will be the host. Does this means they're going to win or dominates a lot of games? I think it's worth to know the answer with data.

Product Catalog Streamlit Web App

Year Accomplished

2026

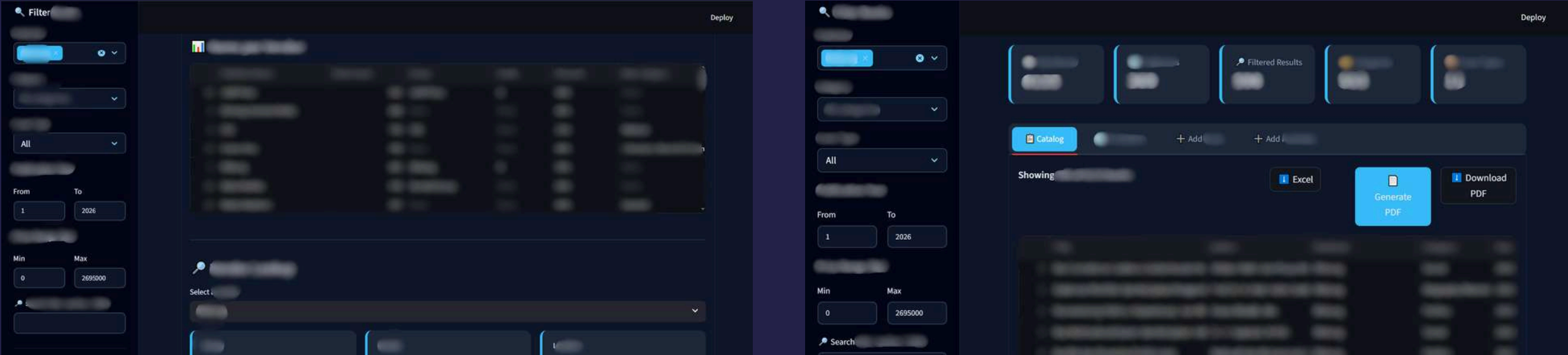
Role/Position

Developer

Project Type

Work Project

This is an interactive web application to manage and explore product catalog data from Excel. I built this with Python and Streamlit that replaces a manual Excel product catalog workflow, resulting in faster filtering, cleaner data management, and standardized PDF file export.



What I Learned

- The importance of having consistent data because real life data are messy
- Debugging is a helpful process and errors are good because it gives you the room for improvement
- Collaboration helps me understand that features are only valuable if they solve a specific daily problems.
- Disagreement with someone who knows better than you is not a bad thing, they're an important point to solve a problem.

How?

When I'm building the web app, I'm offering a suggestion to a co-worker. In my point of view some of the columns looks redundant. For efficiency I thought this looks messy and unstructured so it would make sense to remove them, but my co-worker insists on keeping them.

The challenges were instead of being annoyed, I paused and asked them to walk me through to understand how the table is used on a daily basis. What he showed me completely changed the situation. Those "redundant" columns were how they cross-referenced a product across different suppliers and categories, it is a little bit messy but an essential part of their workflow that was completely ignored by me from the technical side.

My "cleaner" version would have broken the workflow.

Why?

I came up with this idea because operating in the existing Excel file was so slow, lots of errors, and frustrating to use for my senior. I saw a clear problem and wanted to offer a solution by giving a cleaner interface, faster filtering, and most importantly a standardized PDF export.

REACT STUDENT PORTAL

2023

Role

Frontend Developer

Project Type

Capstone Project – Ruangguru

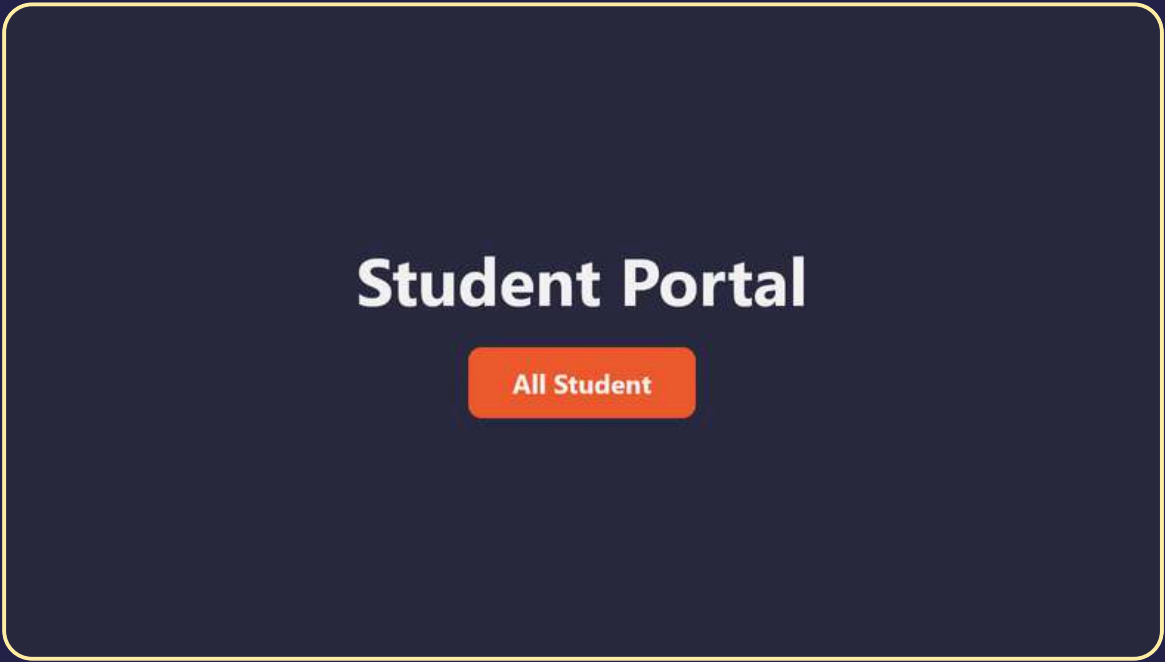
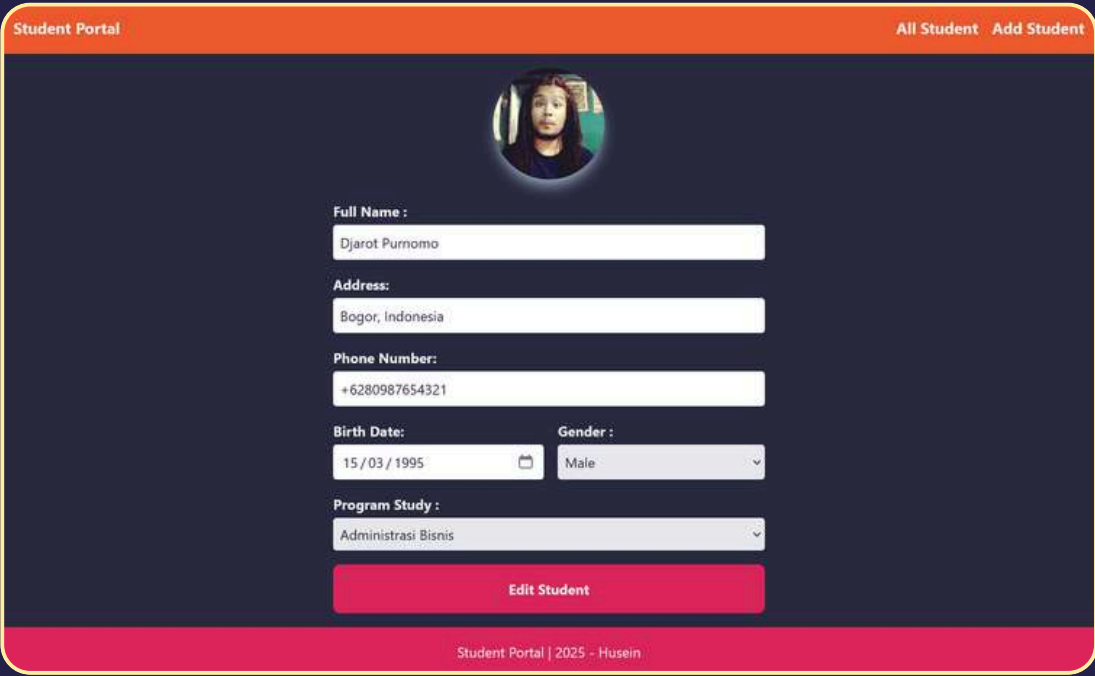
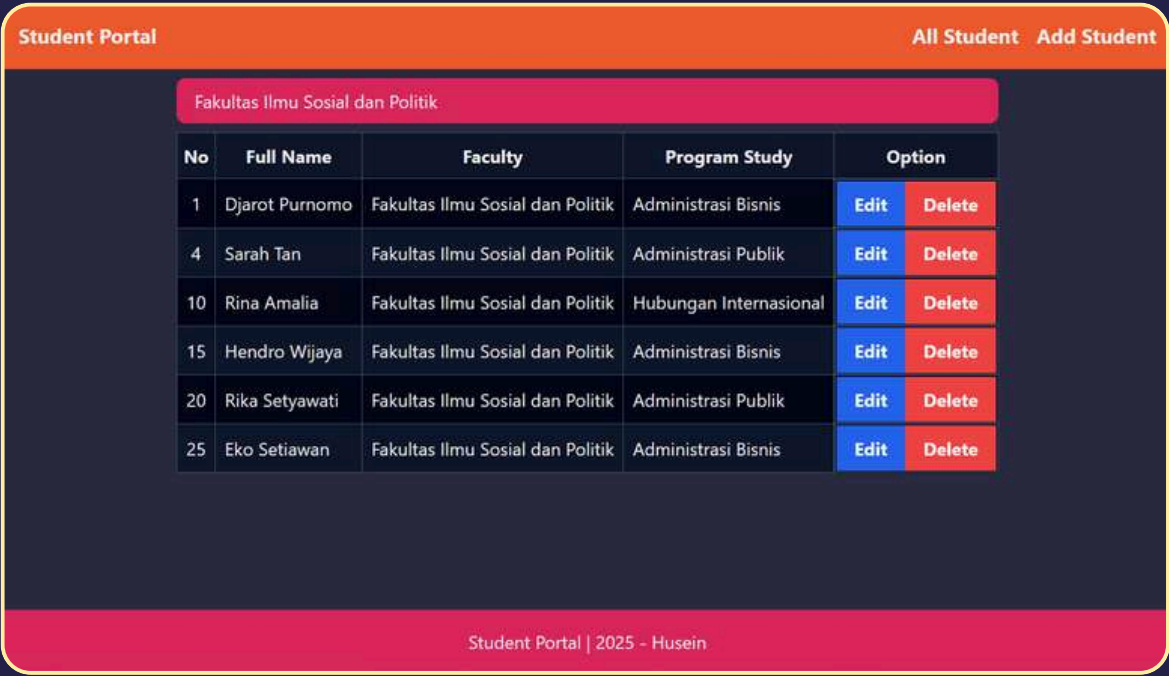
Link

<https://github.com/fedexid/student-portal>

What I learned

- How to implement a CRUD app with React
- Understanding different uses of React Hooks and Router for navigating
- Realized the importance of a responsive website for different devices

A Student Portal Web Application for users or admin to manage students data where they can add, edit, and delete students in the table. It’s a web app with **CRUD** operations developed using **React + React Router** for the page navigation, Tailwind for the responsive styling, and JSON-server for the students data.



Personal Website

A personal website to showcase things about me and my projects, pages include Home, Projects, and About Me section. Its build with React + Typescript, TailwindCSS for the styling, and Framer-Motion for the animation.

Year Accomplished
2026

Role/Position
Designer and Developer

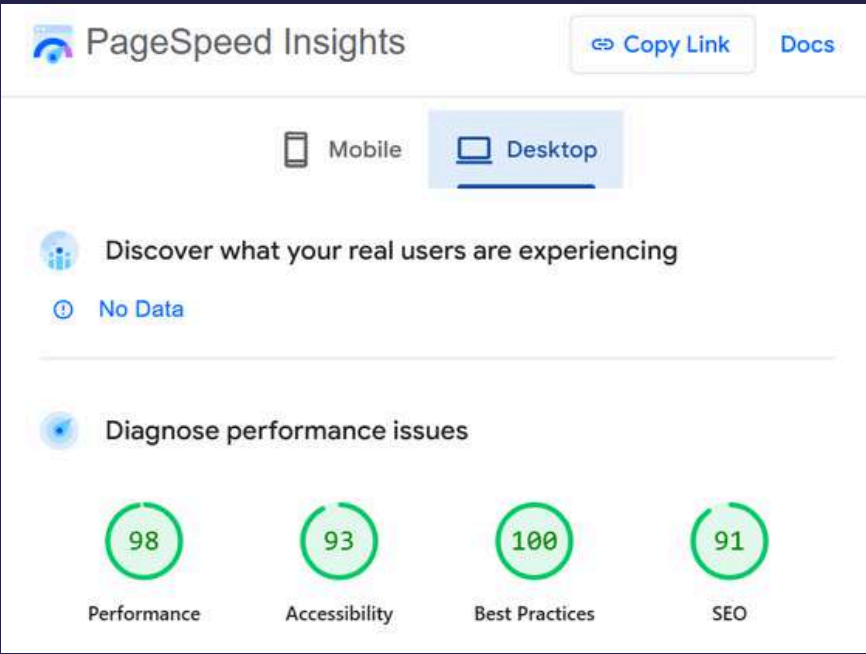
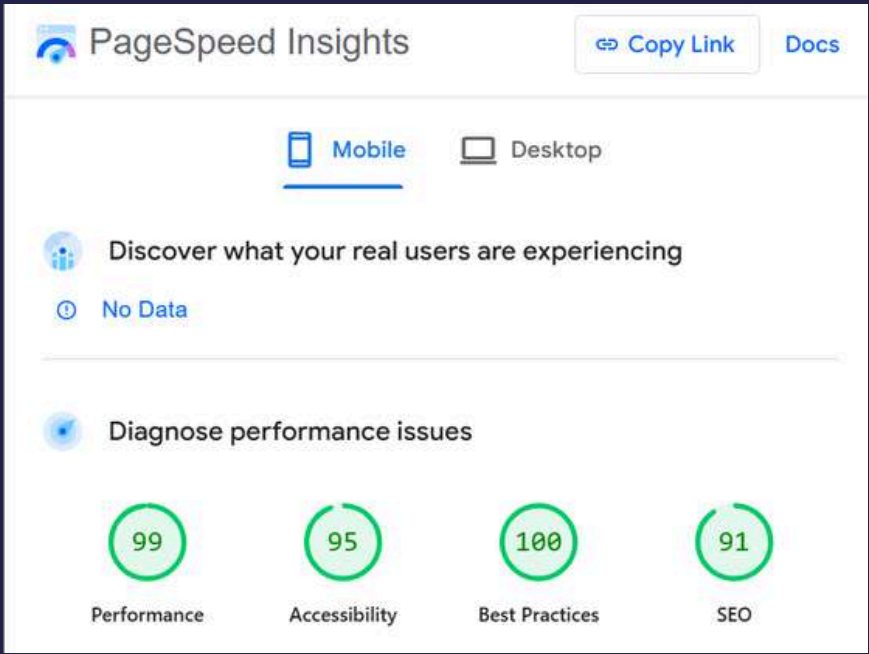
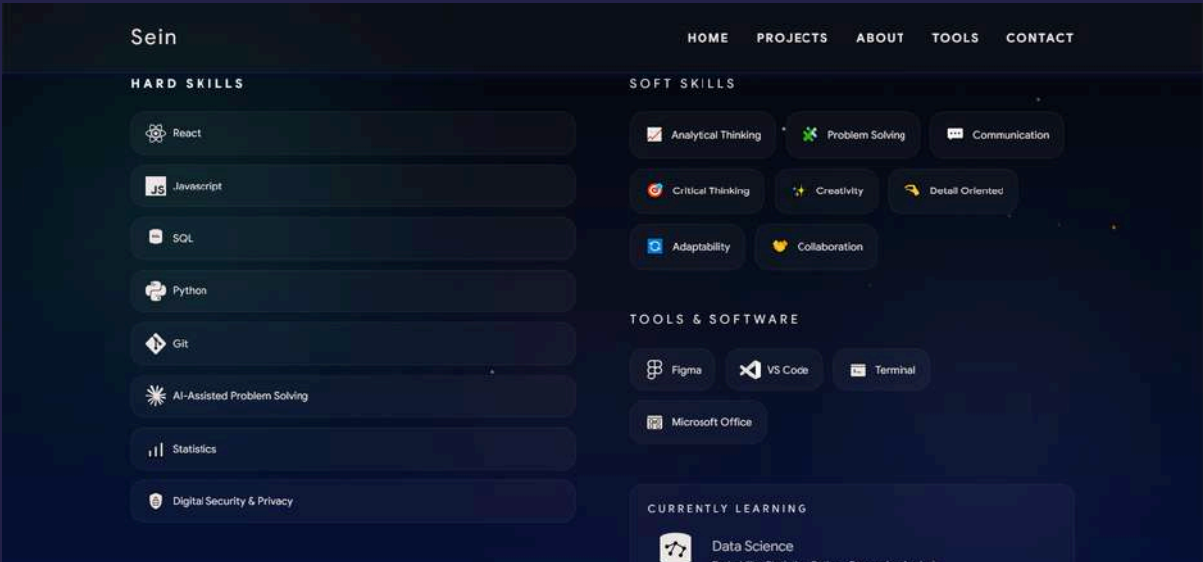
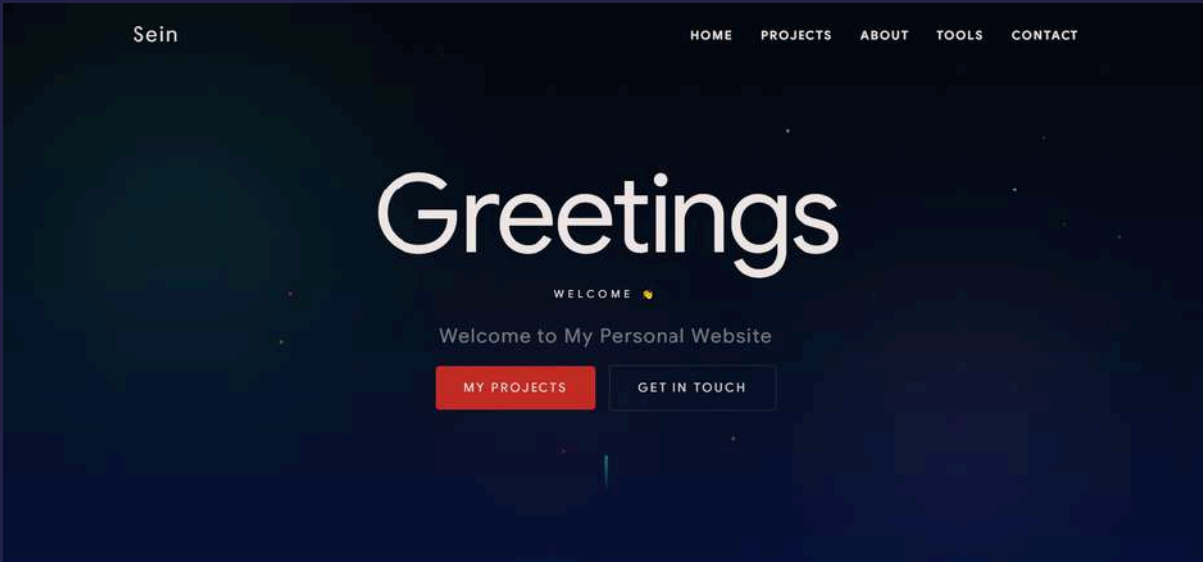
Individual/Group
Individual

Project Type
Self-Initiated

Publication Link

- <https://sein-portfolio.pages.dev/>
- <https://github.com/fedexid/personal-web>

- What I Learn?
- Better understanding of React, Component Structure, and layout design
 - Learn about core concept of Typescript
 - Best approach to make responsive styling and animations
 - Learn about optimization, which led to a good PageSpeed Insight score



My website achieved **98-99%** Pagespeed Insight score for **Performance** and **93-95%** for **Accessibility**, which comes from intentional improvements to ensure a fast and user-friendly experience. Whilst maintaining **100% score for best practice** and **91%** for **Search Engine Optimization (SEO)**

ACE: HOME OF TENNIS

Ongoing

Role

UI/UX Designer

Project Type

Self-Initiated

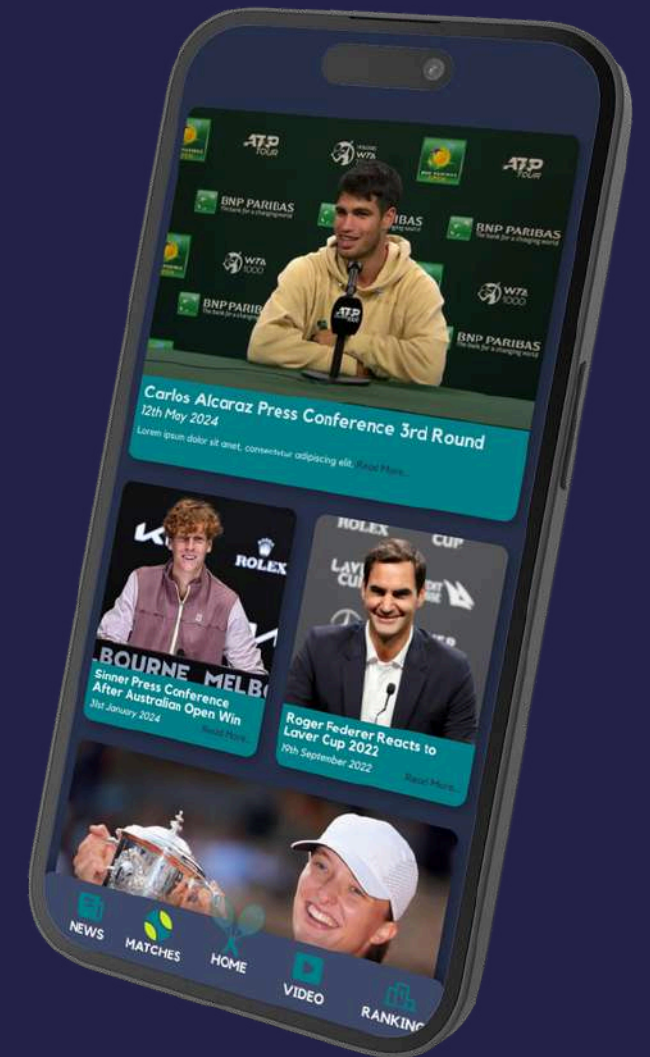
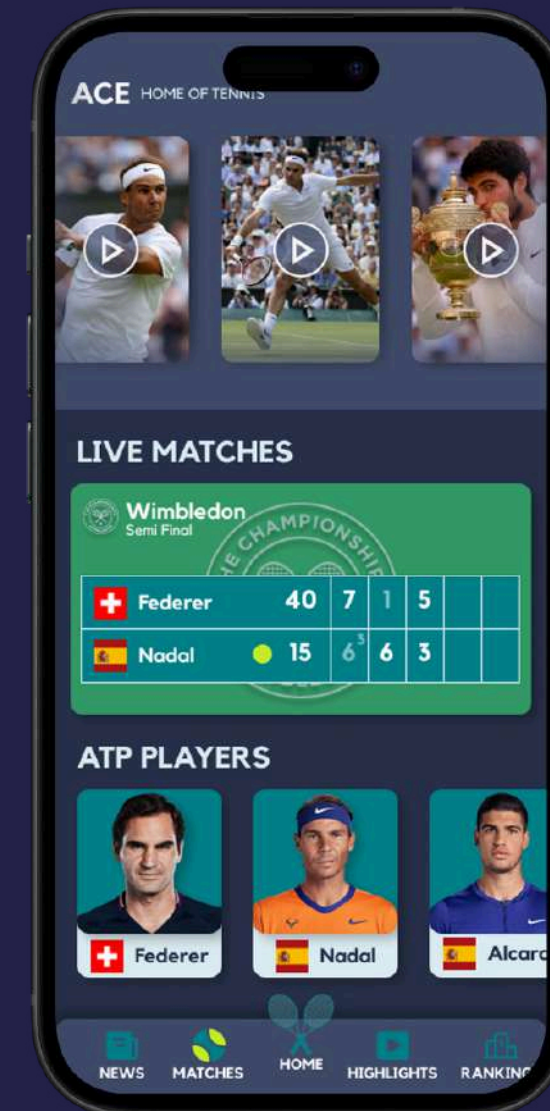
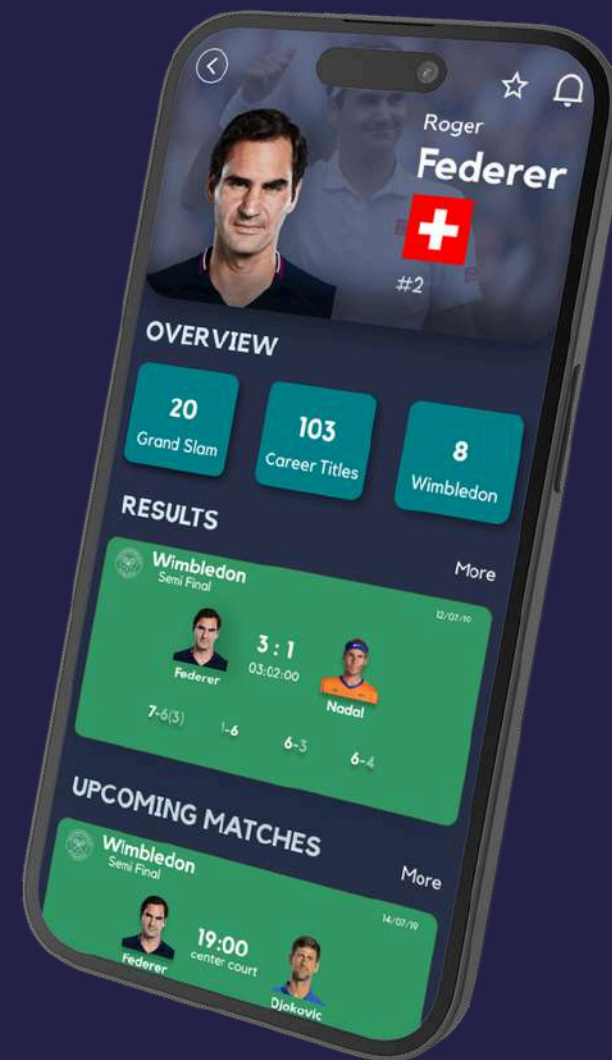
Link

<https://www.figma.com/design/YD2LzcgDA22nY0xcSTwknm/Tennis-App?t=zJ5tI9OM4nai23bO-1>

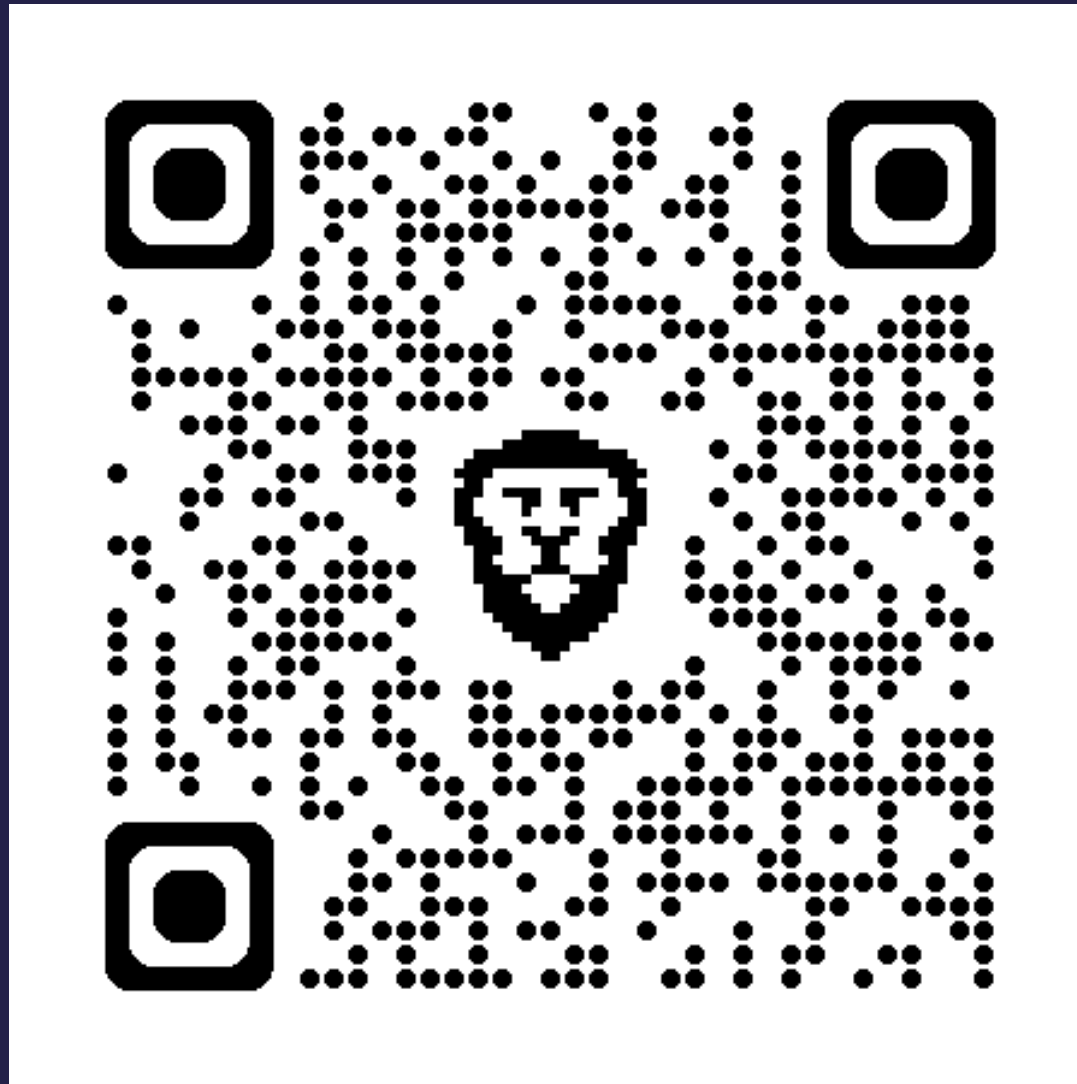
What I learned

- Learn the importance of typography and visual consistency
- Realized the importance of considering user needs and maintaining clarity in design when crafting solutions, even without formal user research.

ACE is an ongoing app made for the home of tennis that gives you every news regarding the world of tennis. This app has many features such as player profile, player rankings, etc. This is still a concept with the focus on giving tennis fans a platform, given the lack of fully tennis based apps for smartphones.



OTHER PROJECTS



<https://drive.google.com/drive/folders/1SsZqvG2emW74tOo5QFkR8h1hbQ5jR6Xv?usp=sharing>